Steps	Reason
1. $\forall x(P(x) \vee Q(x))$	Premise
2. $\forall x((\neg P(x) \wedge Q(x)) \rightarrow R(x))$	Premise
3. $\neg P(a) \wedge Q(a) \rightarrow R(a)$	Universal instantiation on 2
4. $\neg(\neg P(a) \wedge Q(a)) \vee R(a)$	Conditional-disjunction equivalence on 3
5. $(\neg(\neg P(a)) \vee \neg Q(a)) \vee R(a)$	De Morgan's Law
6. $(P(a) \vee \neg Q(a)) \vee R(a)$	Double negation law on 5
7. $(\neg Q(a) \vee P(a)) \vee R(a)$	Commutative law on 6
8. $\neg Q(a) \vee (P(a) \vee R(a))$	Associative law on 7
9. $P(a) \vee Q(a)$	Universal instantiation on 1
10. $Q(a) \vee P(a)$	Commutative law on 9
11. $P(a) \vee (P(a) \vee R(a))$	Resolution on 10,8
12. $(P(a) \vee P(a)) \vee R(a)$	Associative law on 11
13. $P(a) \vee R(a)$	Idempotent law on 12
14. $R(a) \vee P(a)$	Commutative law on 13
15. $\neg(\neg R(a)) \vee P(a)$	Double negation law on 14
16. $\neg R(a) \rightarrow P(a)$	Conditional-disjunction equivalence on 15
17. $\forall x(\neg R(x) \rightarrow P(x))$	Universal generalization on 16

29. Use rules of inference to show that if $\forall x(P(x) \vee Q(x))$, $\forall x(\neg Q(x) \vee S(x))$, $\forall x(R(x) \rightarrow \neg S(x))$, and $\exists x \neg P(x)$ are true, then $\exists x \neg R(x)$ is true.

Soln: Premises: 1. $\forall x(P(x) \vee Q(x))$ 2. $\forall x(\neg Q(x) \vee S(x))$
3. $\forall x(R(x) \rightarrow \neg S(x))$ 4. $\exists x \neg P(x)$

Conclusion: $\exists x \neg R(x)$